EXAMPLE 7.15

Two polaroids are kept with their transmission axes inclined at 30°. Unpolarised light of intensity I falls on the first polaroid. Find out the intensity of light emerging from the second polaroid.

Solution

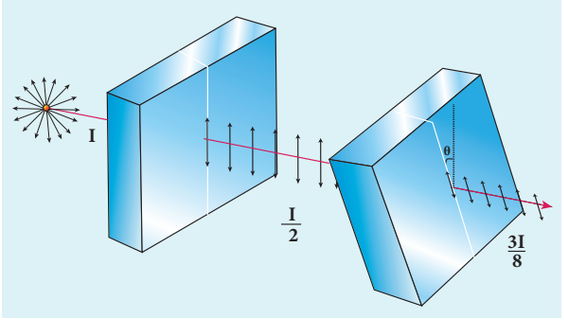
As the intensity of the unpolarised light falling on the first polaroid is I, the intensity of polarized light emerging from it will be, $I_0 = \left(\frac{I}{2}\right)$. Let I' be the intensity of light emerging from the second polaroid.

Malus' law, $I' = I_0 \cos^2\theta$

Substituting,

$$I' = \left(\frac{I}{2}\right)\cos^2(30^\circ) = \left(\frac{I}{2}\right)\left(\frac{\sqrt{3}}{2}\right)^2 = I\frac{3}{8}$$

$$I' = \left(\frac{3}{8}\right)I$$



EXAMPLE 7.16

Two polaroids are kept crossed (transmission axes at 90°) to each other.

(a) What will be the intensity of the light coming out from the second polaroid when an unpolarised light of intensity I falls on the first polaroid?

(b) What will be the intensity of light coming out from the second polaroid if a third polaroid is kept in between at 45° inclination to both of them.

Solution

(a) As the intensity of the unpolarised light falling on the first polaroid is I, the intensity of polarized light emerging from it will be $I_0 = \left(\frac{I}{2}\right)$. Let I' be the intensity of light emerging from the second polaroid.

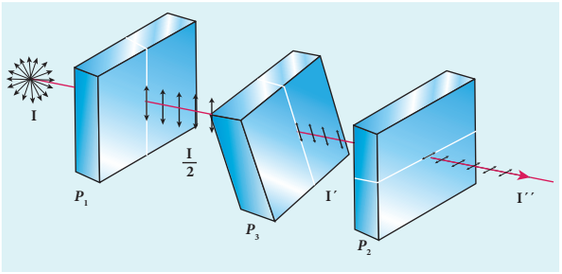
Malus' law, $I' = I_0 \cos^2\theta$

Here θ is 90° as the transmission axes are perpendicular to each other.

Substituting,

$$I' = \left(\frac{I}{2}\right)\cos^2(90^\circ) = 0 \quad \left[\because \cos(90^\circ) = 0\right]$$

No light comes out from the second polaroid.



(b) Let the first polaroid be P_1 and the second polaroid be P_2. They are oriented at 90°. The third polaroid P_3 is introduced between them at 45°. Let I' be the intensity of light emerging from P_3.

Angle between P_1 and P_3 is 45°. The intensity of light coming out from P_3 is, $I' = I_0 \cos^2\theta$

Substituting,

$$I' = \left(\frac{I}{2}\right)\cos^2(45^\circ) = \left(\frac{I}{2}\right)\left(\frac{1}{\sqrt{2}}\right)^2 = \frac{I}{4'};\ I' = \frac{I}{4}$$

Finally, the light has to pass through P_2. Angle between P_3 and P_2 is 45°. Let I'' is the intensity of light coming out from P_2

$I'' = I' \cos^2\theta$

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